

Face Entry Pro

Biometric Attendance & Campus Intelligence System

Software Requirements Specification (SRS)

System Requirements

Software Requirements:

- Operating System: Windows, Linux (Development), Linux (Raspberry Pi), Android, iOS (Mobile Access).
- Languages: Python, JavaScript, HTML, CSS.
- Frameworks: Django / Flask (Backend), Bootstrap (Frontend), REST API for mobile integration.
- Mobile App (Optional): Flutter / React Native (for Android & iOS access).
- AI Libraries: TensorFlow Lite, OpenCV, MediaPipe, Face Recognition (MobileFaceNet).
- Database: MySQL.
- Tools: GitHub, VS Code, Postman.

Functional Requirements

FR01	User Login System
Description	Allows students, teachers, parents, and admins to securely log into the system
Stakeholders	Students, Faculty, Parents, Admin

FR02	User Registration
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Description	Registers new students, faculty, and staff into the system database
Stakeholders	Admin

FR03	Face Enrollment
Description	Captures and stores student facial features during registration
Stakeholders	Admin, Students

FR04	Face Detection
Description	Detects human faces from camera input in real time
Stakeholders	System, Students

FR05	Face Recognition
Description	Identifies students by matching detected faces with stored facial data
Stakeholders	System

FR06	Attendance Marking
Description	Automatically marks attendance after successful face verification
Stakeholders	System, Faculty

FR07	Attendance History View
Description	Allows users to view previous attendance records
Stakeholders	Students, Faculty

FR08	Attendance Report Generation
Description	Generates daily, weekly, monthly, and semester-based reports
Stakeholders	Faculty, Admin

FR09	Real-Time Attendance Monitoring
Description	Displays live attendance status of students
Stakeholders	Faculty, Admin

FR10	Proxy Attendance Prevention
Description	Prevents fake attendance using biometric verification
Stakeholders	System, Admin

FR11	Liveness Detection
Description	Verifies whether the detected face belongs to a real person
Stakeholders	System

FR12	Anti-Spoofing Detection
Description	Prevents attendance using photos, videos, or mobile screens
Stakeholders	System

FR13	Student Profile Management
Description	Allows admin to create and update student information
Stakeholders	Admin

FR14	Faculty Profile Management
Description	Allows management of faculty information and permissions
Stakeholders	Admin

FR15	Faculty Dashboard
Description	Provides attendance analytics and student monitoring features
Stakeholders	Faculty

FR16	Admin Dashboard
Description	Provides complete control over system operations
Stakeholders	Admin

FR17	Student Dashboard
Description	Allows students to view attendance and notifications
Stakeholders	Students

FR18	Parent Monitoring System
Description	Allows parents to monitor student attendance status
Stakeholders	Parents

FR19	Notification Management
Description	Sends alerts for low attendance and important updates
Stakeholders	Students, Parents, Faculty

FR20	Low Attendance Alert
Description	Automatically notifies students with attendance shortages
Stakeholders	Students, Faculty

FR21	Biometric Data Storage
Description	Stores encrypted facial embeddings securely
Stakeholders	System, Admin

FR22	Data Encryption
Description	Protects sensitive biometric and personal information
Stakeholders	System

FR23	Camera Integration
Description	Connects CCTV/webcam devices for face capturing
Stakeholders	System

FR24	Multi-Camera Support
Description	Supports multiple cameras at different campus locations

Stakeholders	Admin, System
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FR25	Smart Video Search
Description	Searches CCTV footage using a person's face image
Stakeholders	Security, Admin

FR26	Incident Detection Logging
Description	Records security incidents such as fights or vandalism
Stakeholders	Security

FR27	Suspicious Activity Detection
Description	Detects unusual behavior and security threats
Stakeholders	System, Security

FR28	System Audit Log
Description	Records user activities and system changes
Stakeholders	Admin

FR29	AI Model Integration
Description	The system integrates AI models for face detection, face recognition, and liveness verification to automate attendance processing.
Stakeholders	System, Developer, Admin

FR30	AI Model Management
Description	Allows updating and maintaining AI recognition models
Stakeholders	Developer, Admin

FR31	Offline Attendance Mode
Description	Allows attendance processing without internet connection

Stakeholders	System
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FR32	Data Synchronization
Description	Synchronizes local Raspberry Pi data with the central database
Stakeholders	System, Admin

FR33	Database Management
Description	Manages student, attendance, and security-related data
Stakeholders	Admin

FR34	Backup and Recovery
Description	Creates backups and restores system data when required
Stakeholders	Admin

FR35	Performance Monitoring
Description	Tracks system speed, accuracy, and resource usage
Stakeholders	Admin, Developer

FR36	System Feedback Collection
Description	The system allows students and faculty to provide feedback about system performance and usability.
Stakeholders	Students, Faculty, Admin

FR37	System Configuration Management
Description	Allows administrators to configure cameras, users, and system settings
Stakeholders	Admin

Non-Functional Requirements

NFR01	Performance
Description	Face recognition should complete within 1–2 seconds per user
Stakeholders	System, Students

NFR02	Reliability
Description	System should operate continuously without failure during working hours
Stakeholders	All Users

NFR03	Security
Description	All authentication and biometric data must be securely encrypted
Stakeholders	System, Admin

NFR04	Safety
Description	System must prevent data loss using automatic backup and recovery
Stakeholders	Admin, System

NFR05	Usability
Description	System interface must be simple and easy to use for all user types
Stakeholders	Students, Faculty

NFR06	Scalability
Description	System must support increasing number of students and multiple cameras
Stakeholders	Admin, System

NFR07	Maintainability
Description	System must be modular so that updates and debugging are easy
Stakeholders	Developer

NFR08	Portability
Description	System must run on Raspberry Pi, web server, and mobile devices
Stakeholders	Users

NFR09	Availability
Description	System must be accessible during campus working hours with minimal downtime
Stakeholders	Admin, Users

NFR10	Accuracy
Description	Face recognition system must maintain high accuracy under normal conditions
Stakeholders	System

NFR11	Interoperability
Description	System must support integration with web and mobile applications via API
Stakeholders	Developer, System

NFR12	Data Integrity
Description	Attendance data must remain consistent and unchanged without authorization
Stakeholders	Admin, System

NFR13	Compliance
Description	System must follow institutional privacy and data protection rules
Stakeholders	Admin

NFR14	Response Time
Description	Attendance marking and system response must occur in real time
Stakeholders	Faculty, System